

FACULTY OF BASIC MEDICAL SCIENCES

MBBS 214: Human Physiology II - Second Semester Examination 2023

Lecturer: Dr. A. Okafor Time allowed: 2 hours Total marks: 40

Instructions: Answer ALL questions in Sections A and B. Answer ONE question in Section C.

SECTION A: Multiple Choice (10 marks). Each question carries 1 mark.

1. Which event produces the first heart sound (S1)?
 - A. Closure of the semilunar valves
 - B. Closure of the atrioventricular valves
 - C. Opening of the atrioventricular valves
 - D. Rapid ventricular filling
2. If end-diastolic volume is 120 mL and end-systolic volume is 50 mL, the stroke volume is:
 - A. 50 mL
 - B. 70 mL
 - C. 120 mL
 - D. 170 mL
3. Ejection fraction is calculated as:
 - A. Stroke volume divided by end-systolic volume
 - B. Stroke volume divided by end-diastolic volume
 - C. End-diastolic volume divided by stroke volume
 - D. Cardiac output divided by heart rate
4. A substance that is freely filtered, neither reabsorbed nor secreted, and whose clearance equals GFR is:
 - A. Glucose
 - B. Para-aminohippuric acid (PAH)
 - C. Inulin
 - D. Urea
5. Glucose is normally absent from urine because it is:
 - A. Not filtered at the glomerulus
 - B. Completely reabsorbed in the proximal tubule
 - C. Secreted into the distal tubule
 - D. Metabolised in the loop of Henle
6. Which of the following opposes glomerular filtration?
 - A. Glomerular capillary hydrostatic pressure
 - B. Plasma colloid osmotic (oncotic) pressure
 - C. High afferent arteriolar flow
 - D. Increased capillary permeability
7. The rapid depolarisation phase of a neuronal action potential is due to:
 - A. K⁺ efflux
 - B. Na⁺ influx through voltage-gated channels

- C. Cl⁻ influx
- D. Ca²⁺ efflux

8. The absolute refractory period exists because:

- A. K⁺ channels are fully open
- B. Voltage-gated Na⁺ channels are inactivated
- C. The Na⁺/K⁺ pump stops working
- D. The membrane potential reaches the Na⁺ equilibrium potential

9. Which lung volume cannot be measured by simple spirometry?

- A. Tidal volume
- B. Vital capacity
- C. Residual volume
- D. Expiratory reserve volume

10. Vital capacity is equal to:

- A. Tidal volume + inspiratory reserve volume
- B. Inspiratory reserve volume + tidal volume + expiratory reserve volume
- C. Expiratory reserve volume + residual volume
- D. Tidal volume + expiratory reserve volume

SECTION B: Short Answer (20 marks). Each question carries 5 marks.

11. Define cardiac output and calculate it for a person with a heart rate of 75 beats per minute and a stroke volume of 70 mL. (5 marks)

12. Explain what is meant by renal clearance. Calculate the clearance of inulin given: plasma inulin 0.25 mg/mL, urine inulin 20 mg/mL, urine flow rate 1.5 mL/min. (5 marks)

13. Describe the phases of a nerve action potential and state the ion movements responsible for each phase. (5 marks)

14. List the four primary lung volumes and state a typical value for each in a healthy adult. (5 marks)

SECTION C: Theory (10 marks). Answer ONE question.

15. Describe the events of the cardiac cycle, with reference to pressure and volume changes in the left ventricle and the timing of the heart sounds. (10 marks)

16. Explain the mechanisms of glomerular filtration and describe how the Starling forces determine the net filtration pressure. (10 marks)